

Unit 3.5. Implicit differentiation

Note Title

2014/03/23

Find the derivative $\frac{dy}{dx}$ if

$$y + \sin y = 2x \quad \text{or}$$

$$x^3 + y^3 = 6xy$$

If we can make y the subject
 $\left[y = \dots x \text{'s} \dots \right]$ then

$\frac{dy}{dx}$ is easy. We say equation is in explicit form. If we cannot make y the subject, then the equation is in implicit form, and we must learn to derive implicitly.

Example:

$$x^2 + y^2 = 9$$

$$x^2 + [f(x)]^2 = 9$$

$$2x + 2[f(x)] \cdot f'(x) = 0$$

$$2x + 2y \cdot y' = 0$$

$$2x + 2y \cdot \frac{dy}{dx} = 0$$

Chain rule.

Examples:

$$\textcircled{1} \quad 2(x+y)^{1/3} = y$$

$$\therefore 2 \cdot \frac{1}{3} (x+y)^{-2/3} (1+y') = y' \quad (y' = ?)$$

$$\frac{2}{3} (x+y)^{-2/3} + \frac{2}{3} (x+y)^{-2/3} \cdot y' = y'$$

$$y' \left[\frac{2}{3} (x+y)^{-2/3} - 1 \right] = - \frac{2}{3} (x+y)^{-2/3}$$

$$y' = \frac{- \frac{2}{3} (x+y)^{-2/3}}{\frac{2}{3} (x+y)^{-2/3} - 1}$$

$$\textcircled{2} \quad \tan(xy) = x+y$$

$$\sec^2(\underline{xy}) \cdot (\underline{1 \cdot y + x \cdot y'}) = 1 + y'$$

$$x y' \sec^2(xy) - y' = 1 - y \sec^2(xy)$$

$$y' = \frac{1 - y \sec^2(xy)}{x \sec^2(xy) - 1}$$

$$\textcircled{3} \quad \frac{x}{y} = x^2 y^3 + 1$$

$$x = x^2 y^4 + y$$

$$\frac{y - x y'}{y^2} = 2x y^3 + x^2 \cdot 3y^2 \cdot y'$$

$$y - \underline{xy'} = 2xy^5 + \underline{x^2 \cdot 3y^4 \cdot y'}$$

$$y'(-x - 3x^2y^4) = 2xy^5 - y$$

$$y' = \dots\dots\dots$$

④ Determine $\boxed{\frac{dy}{dx} \big|_{(1,2)}}$ if

$$\begin{matrix} x=1 \\ y=2 \end{matrix}$$

$$-ye^{4x^2} = 2xe^{y^2}$$

$$y' \cdot e^{4x^2} + y \cdot e^{4x^2} \cdot 8x = 2(e^{y^2} + x \cdot e^{y^2} \cdot 2y \cdot y')$$

$$y' \big|_{(1,2)} = 2 \quad (\text{Check!})$$

Determine $\frac{dy}{dx}$ if $y = \sin^{-1}x$.

$$y = \sin^{-1}x \iff x = \sin y \text{ and } y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\Rightarrow 1 = \cos y \cdot \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$$

$$= \frac{1}{\sqrt{1 - \sin^2 y}}$$

$$= \frac{1}{\sqrt{1 - x^2}}$$

$$\swarrow \cos y > 0$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arcsin[\underline{f(x)}] = \frac{\underline{f'(x)}}{\sqrt{1-[\underline{f(x)}]^2}}$$

Determine $\frac{dy}{dx}$ if $y = \tan^{-1}x$

$$y = \tan^{-1}x \iff \tan y = x \text{ and } y \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$$\Rightarrow \sec^2 y \cdot \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1 + \tan^2 y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1 + x^2}$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \arctan[\underline{f(x)}] = \frac{\underline{f'(x)}}{1 + [\underline{f(x)}]^2}$$

$$\textcircled{1} \quad y = \sqrt{x} \cdot \arcsin(x^2)$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}} \cdot \arcsin(x^2) + \sqrt{x} \cdot \frac{2x}{\sqrt{1-(x^2)^2}}$$

$$\textcircled{2} \quad y = \arcsin(\sin x)$$

$$y' = \frac{\cos x}{\sqrt{1-(\sin x)^2}} = 1$$

$$\textcircled{3} \quad y = \arctan(\cot x)$$

$$y' = \frac{-\operatorname{cosec}^2 x}{1 + \cot^2 x}$$

$$\textcircled{4} \quad y = \arcsin(2^x)$$

$$\frac{dy}{dx} = \frac{2^x \ln 2}{\sqrt{1-(2^x)^2}}$$